

CHIARA LISOTTI

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PERSONAL PROFILE

I have recently submitted my PhD thesis, completed as part of the astroparticle physics group at the University of Sydney. My research involves performing theoretical calculations regarding the directional detection of feebly interacting particles such as neutrinos and dark matter across different sources.

ACADEMIC STUDIES

PhD Candidate in Astroparticle Physics Theory

University of Sydney

January 2023 – Present

Sydney, Australia

First Class Honours B.A. Hons Bachelor Degree in Theoretical Physics

Trinity College Dublin

September 2018 – May 2022

Dublin, Ireland

PAPERS AND PUBLICATIONS

'Discriminating Dark Matter Origins with Directional Detection'

N. F. Bell, C. Lisotti, J. L. Newstead, C. A. J. O'Hare, I. Shaukat Ali

March 11, 2026

[2603.10434]

'Directional recoil detection for $CE\nu NS$ measurements with light nuclei at the Spallation Neutron Source'

C. A. J. O'Hare, A. Chandra Shekar, C. Lisotti, M. Litke, N. Mishra, J. L. Newstead, L. E. Strigari, S. E. Vahsen

February 23, 2026

[2602.19618]

'Searching for beyond-Standard-Model solar neutrino interactions using directional detectors'

A. Chandra Shekar, C. Lisotti, N. Mishra, C. A. J. O'Hare, L. E. Strigari

June 11, 2025

[2506.09322] Phys. Rev. D 112, 123021 (2025)

'CYG ν S: Detecting solar neutrinos with directional gas time projection chambers'

C. Lisotti, C. A. J. O'Hare, E. Baracchini, V. U. Bashu, L. J. Bignell, F. D., M. Ghrear, G. J. Lane, L. J. McKie, P. C. McNamara, S. Torelli

April 2, 2024

[2404.03690] Eur. Phys. J. C 84, 1021 (2024)

BURSARIES

The Edna Briggs Scholarship in Physics

University of Sydney

📅 March 2026

10000 AUD

Special Initiatives Funding

ARC Center of Excellence for Dark Matter

📅 April 2025

Funding amounting to a max of 5000AUD to be used for physics outreach on social media

Postgraduate Research Support Scheme 2025

University of Sydney Faculty of Science

📅 March 2025

PhD Scholarship in Dark Matter Detection

University of Sydney

📅 January 2023

AWARDS

Best Talk Award

2025 ARC Center of Excellence for Dark Matter Annual Meeting

📅 November 2025

Outreach and Impact Award

2025 ARC Center of Excellence for Dark Matter Annual Meeting

📅 November 2025

Collaboration and Centre Values Award

2025 ARC Center of Excellence for Dark Matter Annual Meeting

📅 November 2025

Faculty of Science Postgraduate Research Prize for Leadership

University of Sydney

📅 May 2025

PRESENTATIONS

'CYG ν Oz: What Next?'

CYGNO x CYGNUS-Oz Workshop

📅 February 26, 2026

📍 Kobe University, Japan

'An overview of CYGNUS' reach for dark matter and neutrino searches 3.0'

9th edition of the international CYGNUS Workshop on Directional Recoil Detection

📅 February 23, 2026

📍 Kobe University, Japan

'An overview of CYGNUS' reach for dark matter and neutrino searches 2.0'

9th Sydney Consortium for Particle Physics and Cosmology Meeting

📅 November 27, 2025

📍 University of New South Wales, Australia

'Directional Detection of Boosted Dark Matter'

ARC Centre of Excellence for Dark Matter Annual Workshop

📅 November 20, 2025

📍 Canberra, Australia

'An overview of CYGNUS' reach for dark matter and neutrino searches'

ARC Centre of Excellence for Dark Matter Annual Workshop

📅 November 21, 2024

📍 Wollongong, Australia

'CYG ν S: Detecting solar neutrinos with directional gas time projection chambers'

Dark Matter Centre Fortnightly Meeting

📅 April 18, 2024

📍 Remote

'Detecting Solar Neutrinos with Directional Gas TPCs'

7th Sydney CPPC Meeting

📅 February 9, 2024

📍 University of Sydney

POSTER PRESENTATIONS

'Searching for bSM solar neutrino interactions using directional detectors '

Summer School on Neutrino Physics Beyond the Standard Model

📅 June 29, 2025

📍 Strasbourg, France

'Optimising recoil imaging detectors for extraterrestrial particles'

IDM 2024 Conference

📅 July 10, 2024

📍 L'Aquila, Italy

'Detectability of Dark Matter and Neutrinos in a Gas TPC'

ARC Centre of Excellence for Dark Matter Particle Physics, Annual Workshop

📅 November 10, 2023

📍 Adelaide, Australia

'Detectability of Dark Matter and Neutrinos in a Gas TPC'

HDR Symposium

📅 November 10, 2023

📍 University of Sydney

SUPERVISION EXPERIENCE

Supervisor

Undergraduate research project on recoil imaging detector optimisation

📅 January 2026 – Present

📍 University of Sydney

Cosupervisor

Undergraduate research project on CCSN neutrino detection

 November 2025 – Present

 University of Sydney

TUTORING EXPERIENCE

Tutor, Course Development

Computational Lab for 2nd Year Physics

 July 2024 – Present

 University of Sydney

Tutor

Computational Modelling

 January 2024 – Present

 University of Sydney

Course Development

Astrophysics Computer Lab for 3rd Year Physics

 January 2025 – February 2025

 University of Sydney

Tutor

Astronomy Elective Courses

 August 2023 – November 2024

 University of Sydney

COMMITTEE WORK

Committee member

Mentoring Committee

 August 2023 – Present

 ARC Centre of Excellence for Dark Matter Particle Physics

HDR Student Representative

School of Physics

 2024 – 2025

 University of Sydney

Committee member

Faculty of Science Conference Organising Committee

 May 2024 – September 2024

 University of Sydney

Committee member

HDR Symposium Committee 2023

 August – November 2023

 University of Sydney

OUTREACH AND PUBLIC ENGAGEMENT

‘DMystified’

Social Media Dark Matter and Physics Outreach

📅 April 2025 – Present

📍 Remote

Volunteer

STEMPals Program

📅 March 2024 – Present

📍 Remote

Volunteer

Dark Matter Outreach in Schools

📅 June 2025

📍 Rural NSW

Mentor

Science Extension Program

📅 October 2024 – May 2025

📍 Remote

Volunteer

"National Quantum & Dark Matter Road Trip 2024" DemoDay Event

📅 August 2024

📍 University of Sydney

Volunteer

"National Quantum & Dark Matter Road Trip 2023" DemoDay Event

📅 July 2023

📍 University of Sydney